

SANTOPRENE® 103-40

A hard, black, versatile thermoplastic vulcanizate (TPV) in the thermoplastic elastomer (TPE) family. This material combines good physical properties and chemical resistance for use in a wide range of applications. This grade of Santoprene® TPV is shear-dependent and can be processed on conventional thermoplastics equipment for injection molding, extrusion, blow molding, thermoforming or vacuum forming. It is polyolefin based and recyclable within the manufacturing stream.

Key Features

- UL listed: file #QMFZ2.E80017, Plastics - Component; file #QMFZ8.E80017, Plastics Certified For Canada - Component
- Excellent ozone resistance

Typical mechanical properties

Stress at 100% elongation, perpendicular	9 MPa	ISO 527-1/-2 or ISO 37
Stress at break, perpendicular	20.7 MPa	ISO 527-1/-2 or ISO 37
Elongation at break, perpendicular	610 %	ISO 527-1/-2 or ISO 37
Brittleness Temperature	-52 °C	ASTM D 746
Low temperature brittleness	-52 °C	ISO 812
Shore D hardness, 15s	41	ISO 48-4 / ISO 868
Shore D hardness, 168h at 150°C	4	ISO 48-4 / ISO 868
Compression set at 70°C, 24h	54 %	ISO 815
Compression Set at 125°C, 70h	61 %	ISO 815
Tear strength, normal	64 kN/m	ISO 34-1

Thermal properties

RTI, electrical, 1.5mm	85 °C	UL 746B
RTI, electrical, 3.0mm	85 °C	UL 746B
RTI, strength, 1.5mm	85 °C	UL 746B
RTI, strength, 3.0mm	85 °C	UL 746B

Specific Application Suitability

Detergent resistance	f3	UL 749
Detergent resistance	f4	UL 2157

Flammability

Burning Behav. at 1.5mm nom. thickn.	HB class	UL 94
Thickness tested	1.5 mm	UL 94
UL recognition	yes	UL 94
Burning Behav. at thickness h	HB class	UL 94
Thickness tested	1.00 mm	UL 94
UL recognition	yes	UL 94
Hot Wire Ignition, 1mm	PLC 4 s	UL 746A
Hot Wire Ignition, 1.5mm	PLC 3 s	UL 746A
Hot Wire Ignition, 3mm	PLC 2 s	UL 746A

SANTOPRENE® 103-40

Electrical properties

Relative permittivity, 60Hz	2.6	IEC 62631-2-1
Comparative tracking index, 23 °C	PLC 0 PLC	UL 746A
Arc Resistance Performance Level Category	PLC 5 class	UL 746B
Electric Strength, Short Time, 2mm	32 kV/mm	ASTM D 149
High Amperage Arc Ignition Category, 1.5 mm	PLC 0 class	UL 746A
High Voltage Arc Tracking Rate	PLC 1 mm/min	UL 746A

Physical/Other properties

Density	950 kg/m³	ISO 1183
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Injection

Drying Temperature	82 °C	
Drying Time, Dehumidified Dryer	3 h	
Processing Moisture Content	0.08 %	
Max. regrind level	20 %	
Melt Temperature Optimum	215 °C	Internal
Max. mould temperature	10 - 52 °C	
Vent depth	25 µm	
Back pressure	0.345 - 0.689 MPa	
Injection speed	fast	

Extrusion

Drying Temperature	82 °C
Drying Time, Dehumidified Dryer	3 h
Melt Temperature Range	210 °C

Processing Texts

Processing Notes	Desiccant drying for 3 hours at 80 °C (180 °F) is recommended. Santoprene® TPV has a wide temperature processing window from 175 to 230 °C (350 to 450 °F) and is incompatible with acetal and PVC.
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Other Approvals

Other Approvals	OEM	Specification	Additional Information
	Stellantis - Chrysler	MS-AR-100 FGN	
	Ford	WSD-M2D441-A	
	GM	GMW15813, Type 9	
	Stellantis - PSA Group	PMP 01994_10_00138	
	Mercedes-Benz Group (Daimler)	DBL 5562	

SANTOPRENE® 103-40

Stellantis - FCA Group	55248/02	EMP-140
Stellantis - FCA Group	B63 0300 01378_23_00032	Coolant hose celanese TPV1hose-125
Hyundai	MS220-05, Type G	
Renault	FRM 18-27-134 /---	
Renault	UB02b	PMR2021
Renault	UM09g	PMR2021
VW Group	VW50123	
ZF Group	ZF 7832.050.153	

NOTICE TO USERS: Values shown are based on testing of laboratory test specimens and represent data that fall within the standard range of properties for natural material. These values alone do not represent a sufficient basis for any part design and are not intended for use in establishing maximum, minimum, or ranges of values for specification purposes. Colourants or other additives may cause significant variations in data values. Properties of moulded parts can be influenced by a wide variety of factors including, but not limited to, material selection, additives, part design, processing conditions and environmental exposure. Other than those products expressly identified as medical grade (including by MT® product designation or otherwise), Celanese's products are not intended for use in medical or dental implants. Regardless of any such product designation, any determination of the suitability of a particular material and part design for any use contemplated by the users and the manner of such use is the sole responsibility of the users, who must assure themselves that the material as subsequently processed meets the needs of their particular product or use. To the best of our knowledge, the information contained in this publication is accurate; however, we do not assume any liability whatsoever for the accuracy and completeness of such information. The information contained in this publication should not be construed as a promise or guarantee of specific properties of our products. It is the sole responsibility of the users to investigate whether any existing patents are infringed by the use of the materials mentioned in this publication. Moreover, there is a need to reduce human exposure to many materials to the lowest practical limits in view of possible adverse effects. To the extent that any hazards may have been mentioned in this publication, we neither suggest nor guarantee that such hazards are the only ones that exist. We recommend that persons intending to rely on any recommendation or to use any equipment, processing technique or material mentioned in this publication should satisfy themselves that they can meet all applicable safety and health standards. We strongly recommend that users seek and adhere to the manufacturer's current instructions for handling each material they use, and entrust the handling of such material to adequately trained personnel only. Please call the telephone numbers listed for additional technical information. Call Customer Services for the appropriate Materials Safety Data Sheets (MSDS) before attempting to process our products.

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SANTOPRENE® 103-40

热塑性硫化弹性体 (TPV) 物性表 · 中文翻译版 · 原文：Celanese 2023-11-09

SANTOPRENE® 103-40 是一种硬质黑色多用途热塑性硫化弹性体 (TPV)，属于热塑性弹性体 (TPE) 大类。本材料兼具良好的物理性能与化学耐受性，适用于多种场景。该牌号具有剪切依赖性，可在通用热塑性加工设备上进行注射成型、挤出、吹塑、热成型或真空成型。以聚烯烃为基体，可在生产流程内回收利用 (Polyolefin based, recyclable)。

一、关键特性 (Key Features)

UL 认证：档案号 #QMFZ2.E80017 (塑料元件)；#QMFZ8.E80017 (加拿大认证塑料元件)
优异的臭氧耐受性 (Excellent ozone resistance)

二、典型力学性能 (Typical Mechanical Properties)

性能指标	数值	单位	测试标准
100% 定伸应力 (拉伸方向垂直)	9	MPa	ISO 527-1/-2 或 ISO 37
断裂应力 (拉伸方向垂直)	20.7	MPa	ISO 527-1/-2 或 ISO 37
断裂伸长率 (垂直方向)	610	%	ISO 527-1/-2 或 ISO 37
脆化温度 Brittleness Temperature	-52	° C	ASTM D 746
低温脆化温度 Low temp. brittleness	-52	° C	ISO 812
肖氏 D 硬度 (15s) Shore D hardness	41	—	ISO 48-4 / ISO 868
肖氏 D 硬度 (150° C × 168h 后)	4	—	ISO 48-4 / ISO 868
压缩永久变形 (70° C × 24h) Compression set	54	%	ISO 815
压缩永久变形 (125° C × 70h) Compression set	61	%	ISO 815
撕裂强度 (法向) Tear strength, normal	64	kN/m	ISO 34-1

三、热性能 (Thermal Properties)

RTI 相对热指数· 电气性能, 1.5mm	85	° C	UL 746B
RTI 相对热指数· 电气性能, 3.0mm	85	° C	UL 746B
RTI 相对热指数· 力学性能, 1.5mm	85	° C	UL 746B
RTI 相对热指数· 力学性能, 3.0mm	85	° C	UL 746B

四、燃烧特性 (Flammability)

燃烧等级 (1.5mm 厚度) Burning Behavior	HB 级		UL 94
燃烧等级 (1.0mm 厚度) Burning Behavior	HB 级		UL 94
热丝点燃 Hot Wire Ignition, 1mm	PLC 4	s	UL 746A
热丝点燃 Hot Wire Ignition, 1.5mm	PLC 3	s	UL 746A
热丝点燃 Hot Wire Ignition, 3mm	PLC 2	s	UL 746A
耐洗涤剂性 Detergent resistance	f3 / f4		UL 749 / UL 2157

SANTOPRENE® 103-40

热塑性硫化弹性体 (TPV) 物性表 · 中文翻译版第 2 页

五、电性能 (Electrical Properties)

相对介电常数 (60Hz) Relative permittivity	2.6	—	IEC 62631-2-1
相对漏电痕迹指数 (23° C) CTI	PLC 0		UL 746A
耐弧性能等级 Arc Resistance	PLC 5	class	UL 746B
短时击穿电压 (2mm) Electric Strength	32	kV/mm	ASTM D 149
大电流耐电弧点燃 (1.5mm) HAI	PLC 0	class	UL 746A
高压电弧追踪速率 HVTR	PLC 1	mm/min	UL 746A

六、物理及其他性能 (Physical / Other Properties)

密度 Density	950	kg/m³	ISO 1183
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七、注射成型加工参数 (Injection Molding Processing Parameters)

干燥温度 Drying Temperature	82	° C	
除湿干燥时间 Drying Time (Dehumidified Dryer)	3	h	
加工允许含水率 Processing Moisture Content	0.08	%	
最大回料比例 Max. regrind level	20	%	
最优熔体温度 Melt Temperature Optimum	215	° C	Internal
模具温度范围 Max. mould temperature	10 – 52	° C	
排气槽深度 Vent depth	25	µm	
背压 Back pressure	0.345 – 0.689	MPa	
注射速度 Injection speed	快 (fast)		

八、加工注意事项 (Processing Notes)

- 建议在 80° C (180° F) 下除湿干燥 3 小时 (desiccant drying)。
- 加工温度窗口宽，范围 175 – 230° C (350 – 450° F)。
- 不兼容缩醛树脂 (acetal) 和聚氯乙烯 (PVC)，混料需严格隔离。

九、主机厂 (OEM) 认证规范 (Other Approvals)

Stellantis – Chrysler (克莱斯勒)	MS-AR-100 FGN
Ford (福特)	WSD-M2D441-A
GM (通用)	GMW15813, Type 9
Stellantis – PSA Group (标致雪铁龙)	PMP 01994_10_00138
Mercedes-Benz Group / Daimler (梅赛德斯-奔驰)	DBL 5562
Stellantis – FCA Group (菲亚特克莱斯勒)	55248/02 EMP-140
Hyundai (现代)	MS220-05, Type G
Renault (雷诺)	FRM 18-27-134 / UB02b / UM09g PMR2021
VW Group (大众集团)	VW50123
ZF Group (采埃孚)	ZF 7832.050.153

十、可替代材料推荐 (根据网络搜索整理)

品名 / Grade	品牌 / Brand	特点对比 (与 103-40 相比)	参考价格
Sarlink® 4040	Teknor Apex (美国)	Shore D ~40 TPV 汽车认证系列；美系OEM供应商；含回收料	¥28–35/kg (进口)
Milastomer™ 5030B / 8030B	Mitsui Chemicals 三井化学 (日本)	Shore A 50–80 TPV系列；汽车内外饰；注塑挤出两用；耐候	¥32–40/kg
Geolast® 701-70	Celanese (原 Monsanto)	Santoprene同厂 TPV 姊妹牌号；耐油性更强；Shore A 70对应	¥30–38/kg
瑞安SF-TPV D40黑	上海瑞安/ 昆山首发等国内厂商	Shore D ~40 国产TPV；外饰/密封件用；价格优势明显；认证较	¥18–25/kg (国内)
科莱恩 Renol™ /国内 SCPX-TPV	多家国内改性厂	定制配色、Shore D 35–45段可调；需向厂家索取汽车级认证资料	¥15–22/kg

注：价格均为到厂未税估算，仅供参考；实际以厂商正式报价为准。国产替代材料使用前须经客户（雷诺/GEA）认可。
本表替代材料研究来源：Teknor Apex 官网、Celanese 产品手册、CMPE360 TPV产业综述（2024）、工程师论坛 Eng-Tips。